XINYU SHEN

Gainesville, FL | (321)-888-0808 | shenxinyu@ufl.edu | U.S. Citizen (No sponsorship required)
Portfolio: https://yuki-portfolio-three.vercel.app/ | LinkedIn: https://www.linkedin.com/in/xinyu-shen-uf | GitHub: https://github.com/Syuki116

SUMMARY

Data Science and Economics senior with experience building reproducible analytics workflows, interactive dashboards, statistical models, and decision oriented reporting across healthcare, research, and market analysis projects.

EDUCATION

University of Florida | B.S. in Data Science & Economics | GPA: 3.85 **Expected May 2027**
Gainesville, FL | Dean's List 2023–2025 | Cutler's Award, 2026

EXPERIENCE

Student Researcher, Dr. Brant Research Group at University of Florida **May 2026 - August 2026**

- Executed large scale R simulation workflows on UF HiPerGator to evaluate SMFsim performance across within group and between group settings, multiple alpha thresholds, and alternative preprocessing methods.
- Compared raw, ComBatMat adjusted, and downsampled outputs; organized parameter settings, model results, and reproducibility records to support interpretation and visualization.
- Tested adjusted alpha thresholds and assessed how normalization affected small versus larger signal effects across scenarios.
- Diagnosed filtering and workflow issues when the initial dataset produced too few eligible observations, then adapted the pipeline and documented the revised analytical approach.

Biostatistics & Data Science Research Intern, Florida Atlantic University **June 2025 - August 2025**

- Selected for an NIH funded program in applied biostatistics and data science; cleaned and transformed clinical stroke data in R using imputation, one hot encoding, and SMOTE.
- Built and compared logistic regression, LASSO, decision tree, and random forest models; evaluated performance with confusion matrices, accuracy, precision, recall, and F1 score.
- Developed an interactive R Shiny application with reactive pages and visual comparisons across original and resampled datasets.
- Supported faculty analyses in SAS and SPSS through file imports, merges, recoding, descriptive statistics, crosstabs, regressions, and reporting tables.

PROJECTS

Florida Healthcare Market Opportunity Analysis | Python, SQL, SQLite, Streamlit, Plotly **August 2026**

- Integrated two annual Florida county health releases into a 67 county dataset using standardized FIPS joins, one to one validation, numeric cleaning, and automated duplicate and missing value checks.
- Calculated population growth and standardized demographic, diabetes, provider supply, and income indicators into transparent percentile based opportunity scores.
- Built Balanced, Healthcare Need, and Growth scenarios and identified six counties that remained in the Top 10 across all three strategic assumptions.
- Created a SQLite analytical layer and Streamlit dashboard using SQL CTEs, CASE logic, DENSE_RANK, interactive comparisons, and executive summaries.
- Translated county level business requirements into KPI definitions, scenario rankings, and an interactive dashboard for market screening.
- Repository: https://github.com/Syuki116/florida-healthcare-market-analysis

DataFest 2025, University of Florida **March 2025**

- Collaborated on a rapid turnaround analytics project, cleaning, merging, and validating large housing datasets using JMP, R, and Python; handled missing values, standardized fields, and engineered derived variables tied to pricing and demand.
- Conducted exploratory data analysis (EDA) and statistical modeling (regression, correlation, and segmented comparisons) to identify key predictors and quantify relationships driving housing outcomes.
- Built clear visualizations and reporting outputs (plots, dashboards style summaries, and formatted tables) to communicate patterns in affordability, location effects, and demographic differences across groups.
- Delivered a concise, data driven presentation for judges, translating technical findings into actionable recommendations while clearly stating assumptions, limitations, and potential sources of bias.

Cardiovascular Disease Prediction Machine Learning Model, University of Florida **April 2024**

- Built and compared supervised classification models (Logistic Regression, Decision Tree, Random Forest, and SVM) on a cardiovascular dataset (~70K records) to predict disease risk and support early risk screening use cases.
- Cleaned and preprocessed clinical features (e.g., age, blood pressure, cholesterol, glucose), performed train/test splits, and evaluated model performance using accuracy and other standard classification metrics (e.g., confusion matrix, precision/recall as applicable).
- Tuned tree based models with hyperparameter optimization (e.g., depth, number of trees, split criteria) and developed a simple prediction workflow/tool that outputs risk estimates from user provided health indicators.

LEADERSHIP

Chinese Students and Scholars Association (CSSA), University of Florida **Treasurer, Webmaster | 2023–2026**

- Managed organization finances as Treasurer by preparing budget requests, tracking reimbursements, maintaining transaction records, supporting nonprofit filing responsibilities, and helping oversee bank account transitions and documentation. Supported Student Government funding and reimbursement processes by reviewing SAR requests, organizing receipts and expense documentation, and monitoring event related spending to ensure accurate financial records.
- Maintained the organization website and supported event operations as Webmaster by creating PowerPoint materials, Kahoot activities, signup forms, and online resources for member engagement and event coordination.

SKILLS

Technical: Python, SQL, R, SQLite, Streamlit, Plotly, R Shiny, SAS, SPSS, JMP, Excel, Git/GitHub, C++, MATLAB | **Analytics:** data cleaning, EDA, dashboarding, regression, classification, hypothesis testing, model evaluation | **Languages:** English and Mandarin (native)